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Cupholder and Phone Mounting System for a Golf Bag

Abstract

A cupholder and phone mounting system for use in a golf bag comprising a golf club compartment and golf bag legs. The system comprises a shaft sub-assembly comprising a rigid rod configured to be removably installed into the golf club compartment alongside clubs in that golf bag, and a cupholder part. The cupholder part comprises a slot configured to hold a camera phone at an angle configured to permit recording of a golf shot. A fastener slot forms an angle between the cupholder and the shaft subassembly when connected such that the cupholder remains approximately vertical while the rigid rod of the shaft sub-assembly is inserted into the golf club compartment of the golf bag and the golf bag is stood on the golf bag legs.

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Background/Summary

RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Application No. 63/563,188, filed on Mar. 8, 2024. The entire teachings of the above application are incorporated herein by reference.

BACKGROUND

[0002] Golf is a very popular sport across the world for people of all ages. For those over the age of 21 in the United States, most enjoy alcoholic drinks while walking the course. Such drinks are typically contained in standard-size and slim-size aluminum cans. While golf carts and push carts have cupholders, golfers who carry their bag are left with nowhere except the ground to leave their drink during each golf shot they take. Repeatedly bending down is not comfortable, and the drink may tip over on unstable grass, get dirty on the bottom of the drink container, or be forgotten since it is out of sight.

[0003] While a previous product is capable of being installed onto the leg of a golf bag and holding a can, its design lacks security of the drink and robustness of the design. It requires the tightening of a screw against the leg of the bag, which can fail over time. It is also positioned at a low position of the bag, which means that bending over for the beverage and reaching below the golf clubs is still required.

[0004] In addition to enjoying drinks while golfing, golf players occasionally like to have videos of their golf shots taken, either to work on improving their swing, to post on social media, or just to capture a scenic moment. These videos are best filmed from directly behind the golfer. Currently, the only way to get a video like this is to try to lean a phone against something, or to ask another person to take the video. Friends are not always around or willing to take video of any shot the golfer desires to be filmed. There are several unipod/tripod products on the market for golf video recording, but those require some setup time in order to take a video, and the average golfer does not want to put in that much effort to take a quick video.

SUMMARY

[0005] A need exists for a cupholder mounting system that is aesthetic, robust, and securely holds a can in a convenient position. A need also exists for a proper phone holder that allows golfers to effortlessly record videos of their own golf shots whenever they please. This can help golfers improve their golf skills and help them capture more special moments.

[0006] An embodiment according to the invention relates to a cupholder and phone mounting system for use in a golf bag. The system may comprise a shaft sub-assembly and a cupholder part. The shaft sub-assembly may comprise three aluminum rods, totaling a length of approximately 41 inches. These may be threaded and connected with set screws. The purpose of breaking this into three parts is to lessen the burden of packaging and shipping a part that is longer, although one solid rod can be used if shipping is not required. The cupholder part is made of plastic and is connected directly to the shaft sub-assembly. The cupholder remains approximately vertical while the golf bag is stood on its legs. The cupholder may be 3D printed, molded with polyurethane resin, injection molded, or machined. The cupholder is capable of holding a regular or slim can, and it also is slotted to allow for camera phones to record video of golf shots from a consistent angle. It also features slots to hold a pencil and tee, both of which only require use once per hole and are undesirable to keep in a pocket. Even if the golfer is not using the cupholder, the invention is light and does not take up much space, so its phone, tec, and pencil holding capabilities can be utilized throughout a round of golf.

[0007] A further embodiment according to the invention comprises a cupholder and phone mounting system comprising a cupholder that is fastened to a shaft, wherein said shaft is configured to be removably installed into a golf bag alongside the clubs of that golf bag. The

cupholder can include slots for a camera phone, golf tee, and pencil.

[0008] In one embodiment, there is provided a cupholder and phone mounting system for use in a golf bag comprising a golf club compartment and golf bag legs. The system comprises a shaft sub-assembly comprising a rigid rod configured to be removably installed into the golf club compartment alongside clubs in that golf bag; and a cupholder part. The cupholder part comprises a cupholder comprising a slot configured to hold a camera phone at an angle configured to permit recording of a golf shot while the rigid rod of the shaft sub-assembly is inserted into the golf club compartment of the golf bag and the golf bag is stood on the golf bag legs. A fastener slot of the cupholder part is configured to permit the cupholder to connect directly to the shaft sub-assembly. The fastener slot forms an angle between the cupholder and the shaft sub-assembly when connected such that the cupholder remains approximately vertical while the rigid rod of the shaft sub-assembly is inserted into the golf club compartment of the golf bag and the golf bag is stood on the golf bag legs.

[0009] In further, related embodiments, the rigid rod may comprise a plurality of aluminum rods assembled together using threading and set screws. The cupholder may comprise a plastic cupholder, which may, for example, be 3D printed, molded with polyurethane resin, injection molded, or machined. The cupholder may be configured to hold at least one of a standard-size drink can and a slim-size drink can; and may comprise features of a size configured to hold each of a standard size drink can and a slim-size drink can. The cupholder may comprise at least one slot configured to hold a golf pencil or a golf tee; and may comprise both a slot configured to hold a golf pencil and a slot configured to hold a golf tee. The system may further comprise a mounting bracket within the fastener slot.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The foregoing will be apparent from the following more particular description of example embodiments, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating embodiments.

[0011] FIG. 1 is an exploded view of a full assembly in accordance with an embodiment of the invention, with a close-up of the cupholder part.

[0012] FIG. 2 depicts a detailed drawing of a cupholder part design in accordance with an embodiment of the invention.

[0013] FIG. 3 depicts a drawing of a shaft sub-assembly in accordance with an embodiment of the invention.

[0014] FIG. 4 depicts a broken-down model of an assembly in accordance with an embodiment of the invention.

[0015] FIG. 5 depicts a close up of the cupholder part in accordance with an embodiment of the invention, with a pencil and tee occupying the slots that are dedicated to each of them.

[0016] FIG. 6 depicts a full assembly in accordance with an embodiment of the invention standing, not installed into a bag.

[0017] FIG. 7 depicts a full assembly in accordance with an embodiment of the invention installed into the bag (from left to right): while empty, holding a camera phone, holding a slim can, and holding a regular can.

[0018] FIG. 8 depicts a view from where a golfer would be using the cupholder in accordance with an embodiment of the invention, with a can installed.

[0019] FIG. 9 depicts a view from where a golfer would be using the cupholder in accordance with an embodiment of the invention, with the cupholder empty.

[0020] FIG. **10** depicts a view of the set screw (left) that is installed into each end of the middle rod, as well as a temporary cover (middle) that is used to protect a thread sealer, and a view of the end of a rod without set screw installed (right), in accordance with an embodiment of the invention.

[0021] FIG. **11** depicts the shaft resting in the golf bag, in accordance with an embodiment of the invention, similar to how regular golf clubs rest.

[0022] FIG. **12** depicts a mounting bracket that can be included within a fastener slot to provide structural support, in accordance with another embodiment of the invention.

DETAILED DESCRIPTION

[0023] A description of example embodiments follows.

[0024] An embodiment according to the invention comprises two parts: the shaft sub-assembly and the cupholder part. The shaft sub-assembly is specifically shown in FIG. **3**. The cupholder part is specifically shown in the top-right of FIG. **1**, and in FIG. **5**.

[0025] FIG. **1** is an exploded view of a full assembly in accordance with an embodiment of the invention. The parts include a cupholder part **101**, shown in close-up 105 in the upper right of FIG. **1**. A screw **106** can, for example, be a tamper-resistant Torx flat head screw (McMaster catalog number 91870A199). A first shaft sub-part **107** can, for example, be 12 inches in length. Two further screws **108** and **109** can, for example, be alloy steel cup-point set screws (McMaster catalog number 91375A176). Two further shaft sub-parts **110** and **111** can, for example, be 14.4 inches in length. A heavy duty round cap **112** (such as McMaster catalog number 9567K56) can be used on the end of the shaft sub-assembly **115**. The shaft sub-parts **107**, **110**, and **111** can be a plurality of aluminum rods that are assembled together using threading and set screws to form a rigid rod **329** (see FIG. **3**) of the shaft sub-assembly **115**.

[0026] FIG. **2** depicts a detailed drawing of a cupholder part design in accordance with an embodiment of the invention.

[0027] An isometric view of the cupholder part **216** is in the upper right of FIG. **2**. The cupholder part **216** can be a plastic cupholder, which can, for example, be 3D printed, molded with polyurethane resin, injection molded, or machined.

[0028] In the upper left of FIG. **2**, there is shown that a portion of the cupholder part **216** is a cupholder that is configured to hold a standard-size drink can, for example by having a radius **217** of about 33.4 mm, while another portion of the cupholder part **216** is configured to hold a slim-size drink can, for example by having a radius **218** of about 29.1 mm. The cupholder part **216** includes a phone slot **219** configured to hold a camera phone at an angle that is configured to permit recording of a golf shot while the rigid rod of the shaft sub-assembly is inserted into the golf club compartment of the golf bag and the golf bag is stood on the golf bag legs. The phone slot **219** can, for example, have a width **228** equal to that needed to hold a typical maximum mobile phone thickness, such as about 12.7 mm. In addition, the cupholder part **216** includes a tee slot **220** configured to hold a golf tee, which can, for example, have a radius of about 2.75 mm; and includes a pencil slot **221** configured to hold a golf pencil, with an example radius of about 3.9 mm.

[0029] In the lower left of FIG. **2**, there is shown that the cupholder part **216** can include a flared opening **222** for case of use for insertion of a drink can. The width **223** across the phone slot can be sized to be equal to a maximum mobile phone width, such as about 85.09 mm. In addition, the phone slot can have multiple different widths **224**, to allow multiple possible different widths of mobile phone to be held snug.

[0030] In the lower right of FIG. **2**, there is shown that the cupholder **216** can include a fastener slot **225** configured to permit the cupholder to connect directly to the shaft sub-assembly **115** (see FIG. **1**), forming an angle **226** between the cupholder **216** and the shaft sub-assembly **115** when connected such that the cupholder **216** remains approximately vertical while the rigid rod **329** (see FIG. **3**) of the shaft sub-assembly **115** is inserted into the golf club compartment of the golf bag and the golf bag is stood on the golf bag legs. For example, the angle **226** may be 45 degrees. A screw **106** (such as an 8-32 screw) can, for example, mount the shaft sub-assembly **115** (see FIG. **1**) at the

angle **226**. The height **227** of the cupholder **216** may, for example, be 107.7 mm.

[0031] FIG. **3** depicts a drawing of a shaft sub-assembly in accordance with an embodiment of the invention. The assembled rigid rod **329** is formed from the three shaft sub-parts **107**, **110**, and **111**. The first shaft sub-part **107** is the top rod, which attached to the cupholder **216** (of FIG. **2**); the second shaft sub-part **110** is the middle rod; and the third shaft sub-part **111** is the bottom rod. Example dimensions of the shaft sub-parts are 304.8 mm for part **107**, 365.76 mm for part **110**, and 365.76 for the part **111**. The heavy-duty cap **112** is to protect the bottom of the golf bag. In a close-up view **330** of the joint between shaft sub-parts (such as between **107** and **110**, and between **110** and **111**), it can be seen that an 8-32 screw **108** (or **109** of FIG. **1**) can be inserted at a depth such as 0.75 inches. Loctite® (sold by Henkel North American Consumer Goods of Rocky Hill, CT, U.S.A.) (or other adhesive) applied via stick can be activated after the user assembles the sub-parts **107/110/111** to form the rigid rod **329**. Liquid Loctite® (or other adhesive) can be applied to the screw **108** for permanence for the middle rod.

[0032] To make the shaft sub-assembly in accordance with an embodiment of the invention, a 72" aluminum rod is cut down using a chop saw to two different sizes: 14.4" and 12". When a 72" rod is cut equally into five pieces, this results in 14.4" pieces with zero waste. When a 72" rod is cut equally into six pieces, this results in 12" pieces with zero waste. The blades must be intended for cutting metal, and the rods must be chamfered to remove the sharp edges that result from the chop saw. Two of the 14.4" rods **110** and **111** (see FIG. **1**), and one of the 12" rods **107** are used for the assembly. An 8-32 tap is used to tap threads **108**, **109** into the ends of each of the rod pieces. One end of one of the 14.4" rods can be left untapped (such as for heavy duty cap **112**). Aluminum shavings must be cleared out of the rods. The 14.4" rod **111** with one end untapped is the bottom rod, and the untapped end is covered with a heavy-duty cap **112**. The other 14.4" rod **110** will have an 8-32 alloy-steel set screw **108**, **109** installed with a thread sealer into each end, exposing about ½" of the 1¼" screws on each side. These screws **108**, **109** are intended to be permanently fastened to this middle rod. This will serve as the middle rod **110**, and it will connect to the bottom **111** and top rods **107**. A thread sealer stick is used to apply a dry thread sealer to the exposed ends of the set screws **108**, **109**, so the shaft will be permanently assembled by the user. A red cap temporarily covers these ends to protect the thread sealer. This is done to both ends of the middle rod, and an example of one end is shown in FIG. **10**, which depicts a view of the set screw (left) that is installed into each end of the middle rod, as well as the temporary cover (middle) that is used to protect the thread sealer, and a view of the end of a rod without set screw installed (right). For packaging, these three rods can be left unconnected. An example of this is shown in FIG. **4**, which depicts a broken-down model of an assembly in accordance with an embodiment of the invention. Connecting the rod in the middle with two exposed screws to each of the other two rods completes the assembly. The top rod is the 12" rod (see **107** in FIG. **1**) and will be fastened to the cupholder part.

[0033] To make the cupholder part in accordance with an embodiment of the invention, the cupholder part **216**, shown in FIG. **2**, may be produced in a number of ways. This part **216** can be 3D printed with PLA, ABS, ASA, or a nylon filament with no supports required. This part can also be cast with polyurethane from a silicone mold, cast from an injection mold, machined, or use another additive manufacturing technique. A stainless steel 8-32 flat screw **106** (see lower right of FIG. **2**) goes through the cupholder part **216** and connects the cupholder part to the 12" top rod **107** (see FIG. **1**) of the shaft sub-assembly. The part **216** is printed or molded with a nesting spot for the screw **106**, allowing it to hold the cupholder part against the shaft sub-assembly at the desired 45-degree angle **226**. A thread sealer is used to make this fastener permanent. A sticker may be applied to the top of the rod for aesthetic and branding purposes. Connecting all three rods using the threads of the middle rod **110** completes the making of full assembly.

[0034] A summary of manufacturing of an embodiment according to the invention is as follows:

[0035] 1) Cut two pieces **110**, **111** of aluminum rod to approximately 14.4" and chamfer each end

[0036] 2) Tap threads into three of the four ends of those two parts **110**, **111** with an 8-32 tap [0037] 3) Cover the untapped end of the rod piece with a heavy-duty cap **112** [0038] 4) Thread 8-32 set screws into both ends of another aluminum rod piece **110** with thread sealer, leaving approximately $\frac{1}{2}$ " of the screws exposed [0039] 5) Apply a dry thread sealer to the exposed parts of the screws [0040] 6) Cut one piece of aluminum rod **107** to approximately 12" and chamfer each end [0041] 7) Tap threads into both ends of that part with an 8-32 tap [0042] 8) 3D print or mold the cupholder part **216** [0043] 9) Connect the cupholder part **216** to one end of the 12" rod **107** with an 8-32 flat screw **106** and thread sealer [0044] 10) Screw the rod **110** with exposed screws into each of the other rods **107** and **111**.

[0045] To use an embodiment according to the invention, the simple assembly of screwing the top **107** and bottom **111** rods to the middle rod **110** is required. An example of a full assembly is shown in FIG. 6, which depicts a full assembly in accordance with an embodiment of the invention standing, not installed into a bag. This assembly is now ready to go into the golf bag, which it does just like any golf club. It is recommended to put it into the top row of the golf bag to take advantage of the positioning that provides above of the iron clubs of the bag, as shown in FIG. 7, which depicts a full assembly in accordance with an embodiment of the invention installed into the bag (from left to right): while empty, holding a camera phone, holding a slim can, and holding a regular can. If the golfer does not desire having the product in their bag for a specific round (perhaps because they are using a golf cart or because they are not having any drinks,) the invention can quickly be removed and left in a car. However, even a golfer who is not planning on having any drinks may desire to use the invention to hold their phone, pencil, and/or tec.

[0046] A cupholder in accordance with an embodiment of the invention is intended to be used while the bag is stood and stationary. This typically occurs when golfers are waiting to tee off, waiting for the greens to clear, or while hitting their golf shot. During this time, the cupholder is there to hold a regular or slim aluminum can, both of which are shown in FIG. 7. To use the cupholder, a can is simply lowered into the cupholder, and slim cans will go slightly deeper and catch a smaller diameter that holds it snug. Regular cans will not go so deep, but will be held snug at a higher position. As a result, the tops of both these cans will be at approximately the same height. The golfer can enjoy some time with their hands free, or they can grab a club just inches away from where they left their drink. Once a golfer is ready to pick up their bag, they can put their club away and grab their drink. The view the golfer would have as they look to grab their drink is shown in FIG. 8, which depicts a view from where a golfer would be using the cupholder in accordance with an embodiment of the invention, with a can installed. Typically, the golfer will not pick up the bag and walk while a can is in the cupholder, because the drink will likely get shaken up and splash around the bag; rather, the golfer walks with the drink in their hand. FIG. 9 depicts a view from where a golfer would be using the cupholder in accordance with an embodiment of the invention, with the cupholder empty.

[0047] To use the camera phone holder in accordance with an embodiment of the invention, the golfer will position their golf bag about 10 feet behind their ball, which is typically around where bags are left. The golfer will start a video on their camera phone and slot their camera phone into the cupholder, shown in FIG. 7, such that the golf ball is approximately in the middle of the camera view. While the video is recording, the golfer hits their shot, and a steady video is recorded from a desirable angle. The golfer grabs their phone and stops the video before picking up their bag and walking towards their ball. This utility can be especially helpful on the driving range, because the video could be setup, and many golf shots can be taken while the phone continues to record.

[0048] Additionally, the golfer may put their tee into the smaller of the holes on the back side of the cupholder and put their pencil into the larger of the holes. This is shown in FIG. 5, which depicts a close up of the cupholder part in accordance with an embodiment of the invention, with a pencil and tee occupying the slots that are dedicated to each of them. Both tees and pencils are pointy, and they are only used once per hole. Carrying them in pockets can be uncomfortable and harder to

access with a golf glove on. These items can remain stored in the assembly while walking with the bag.

[0049] FIG. 11 depicts the shaft resting in the golf bag, in accordance with an embodiment of the invention, similar to how regular golf clubs rest.

[0050] In another embodiment, shown in FIG. 12, a mounting bracket 1231 can be included within the fastener slot 225 to provide structural support.

[0051] Several exemplary advantages of an embodiment according to invention are listed below:

[0052] 1) By positioning the center of gravity of a 12-ounce can below the axis of the shaft, it ensures that no torque will be applied when a can is placed into the cupholder. If it were above that axis or to the left or right, it would create a torque that tips the drinks over. [0053] 2) Giving the cupholder its own mounting system was important, because if it were to clip onto a club or headcover, that would require making adjustments to the mounting system throughout a round of golf. [0054] 3) This shaft mounting system universally fits into any bag. [0055] 4) The length of the shaft provides exceptional positioning for the cupholder without getting in the way of the golf clubs that need to be accessed. [0056] 5) The shaft mounting system allows for very quick installation and removal from the bag. [0057] 6) The design of the shaft allows for feasible packaging and shipping, yet quick and easy assembly upon receiving the package. [0058] 7) Material choices are durable and corrosion/rust resistant. [0059] 8) The positioning of the cupholder allows for drinks to be dropped off and grabbed inches away from the clubs that are conversely being grabbed and dropped off. [0060] 9) Prevents excessive bending over to drop off and grab beverage can during each of the ~100 golf shots taken per round. [0061] 10) Accommodates regular and slim cans, which make up a majority of the drinks enjoyed on a golf course. Sports drinks have caps and simply can be stored inside a golf bag because of that. [0062] 11) The bracket connected to the cupholder greatly increases the structural integrity of the cupholder and its connection to the shaft. [0063] 12) Phone camera phone holder allows for golfers to record their own shots, which has never been easy for most golfers. [0064] 13) Allows golfers to improve their swing by being able to easily record any shot and review what they need to improve. [0065] 14) Also holds a tee and pencil, which would otherwise be inside the golfer's pocket throughout a round of golf, or lost inside a zipper.

[0066] The teachings of all patents, published applications and references cited herein are incorporated by reference in their entirety.

[0067] While example embodiments have been particularly shown and described, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the embodiments encompassed by the appended claims.

Claims

1. A cupholder and phone mounting system for use in a golf bag comprising a golf club compartment and golf bag legs, the system comprising: a shaft sub-assembly comprising a rigid rod configured to be removably installed into the golf club compartment alongside clubs in that golf bag; and a cupholder part comprising: (i) a cupholder comprising a slot configured to hold a camera phone at an angle configured to permit recording of a golf shot while the rigid rod of the shaft sub-assembly is inserted into the golf club compartment of the golf bag and the golf bag is stood on the golf bag legs; and (ii) a fastener slot configured to permit the cupholder to connect directly to the shaft sub-assembly, forming an angle between the cupholder and the shaft sub-assembly when connected such that the cupholder remains approximately vertical while the rigid rod of the shaft sub-assembly is inserted into the golf club compartment of the golf bag and the golf bag is stood on the golf bag legs.

2. The system of claim 1, wherein the rigid rod comprises a plurality of aluminum rods assembled together using threading and set screws.

3. The system of claim 1, wherein the cupholder comprises a plastic cupholder.
 4. The system of claim 3, wherein the cupholder is 3D printed, molded with polyurethane resin, injection molded, or machined.
 5. The system of claim 1, wherein the cupholder is configured to hold at least one of a standard-size drink can and a slim-size drink can.
 6. The system of claim 5, wherein the cupholder comprises features of a size configured to hold each of a standard-size drink can and a slim-size drink can.
 7. The system of claim 1, wherein the cupholder further comprises at least one slot configured to hold a golf pencil or a golf tee.
 8. The system of claim 7, wherein the cupholder comprises both a slot configured to hold a golf pencil and a slot configured to hold a golf tee.
 9. The system of claim 1, further comprising a mounting bracket within the fastener slot.
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